APEC 8221 - Assignment 1: Tidyverse Fundamentals

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2025-09-19

GitHub Repository URL: [https://github.com/Runcheng112/apec8221-xu001296.git]

1 Setup

```r
# Load required packages
library(tidyverse)
library(gapminder)

# Check data structure
glimpse(gapminder)
```

```
Rows: 1,704
Columns: 6
$ country   <fct> "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", ~
$ continent <fct> Asia, Asia, Asia, Asia, Asia, Asia, Asia, Asia, Asia, Asia, ~
$ year      <int> 1952, 1957, 1962, 1967, 1972, 1977, 1982, 1987, 1992, 1997, ~
$ lifeExp   <dbl> 28.801, 30.332, 31.997, 34.020, 36.088, 38.438, 39.854, 40.8~
$ pop       <int> 8425333, 9240934, 10267083, 11537966, 13079460, 14880372, 12~
$ gdpPercap <dbl> 779.4453, 820.8530, 853.1007, 836.1971, 739.9811, 786.1134, ~
```

2 Data Analysis & Visualization Practice

2.1 Top Performers Identification

Find the top 3 countries in each continent for highest life expectancy and highest GDP per capita in 2007.

2.1.1 Highest Life Expectancy by Continent (2007)

```r
library(dplyr)

# Top 3 by Life Expectancy
top_lifeExp <- gapminder %>%
  filter(year == 2007) %>%
  group_by(continent) %>%
  arrange(desc(lifeExp), .by_group = TRUE) %>%
  slice_head(n = 3) %>%
  select(continent, country, lifeExp)

#
print(top_lifeExp, n = Inf)
```

```
# A tibble: 14 x 3
# Groups:   continent [5]
   continent country          lifeExp
   <fct>     <fct>              <dbl>
 1 Africa    Reunion             76.4
 2 Africa    Libya               74.0
 3 Africa    Tunisia             73.9
 4 Americas  Canada              80.7
 5 Americas  Costa Rica          78.8
 6 Americas  Puerto Rico         78.7
 7 Asia      Japan               82.6
 8 Asia      Hong Kong, China    82.2
 9 Asia      Israel              80.7
10 Europe    Iceland             81.8
11 Europe    Switzerland         81.7
12 Europe    Spain               80.9
13 Oceania   Australia           81.2
14 Oceania   New Zealand         80.2
```

```r
# Your code here - find top 3 countries by life expectancy in each continent
# Remember to use group_by(), arrange(), and slice_head()
```

In 2007, countries in Europe and Asia showed the highest life expectancies, with places like Hong Kong, Japan, and Iceland exceeding 82 years.

In contrast, African countries such as Réunion, Libya, and Tunisia had much lower life expectancies, around 76 years, highlighting the health and living standard disparities across continents.

2.1.2 Highest GDP per Capita by Continent (2007)

```r
# Top 3 by GDP per capita (2007)
top_gdpPercap <- gapminder %>%
  filter(year == 2007) %>%
  group_by(continent) %>%
  arrange(desc(gdpPercap), .by_group = TRUE) %>%
  slice_head(n = 3) %>%
  select(continent, country, gdpPercap)

print(top_gdpPercap, n = Inf)
```

```
# A tibble: 14 x 3
# Groups:   continent [5]
   continent country            gdpPercap
   <fct>     <fct>                  <dbl>
 1 Africa    Gabon                 13206.
 2 Africa    Botswana              12570.
 3 Africa    Equatorial Guinea     12154.
 4 Americas  United States         42952.
 5 Americas  Canada                36319.
 6 Americas  Puerto Rico           19329.
 7 Asia      Kuwait                47307.
 8 Asia      Singapore             47143.
 9 Asia      Hong Kong, China      39725.
10 Europe    Norway                49357.
11 Europe    Ireland               40676.
12 Europe    Switzerland           37506.
13 Oceania   Australia             34435.
14 Oceania   New Zealand           25185.
```

```r
# Your code here - find top 3 countries by GDP per capita in each continent
```

In 2007, the highest GDP per capita countries were mostly in Europe, Asia, and the Americas, with Norway, Kuwait, and the United States leading.

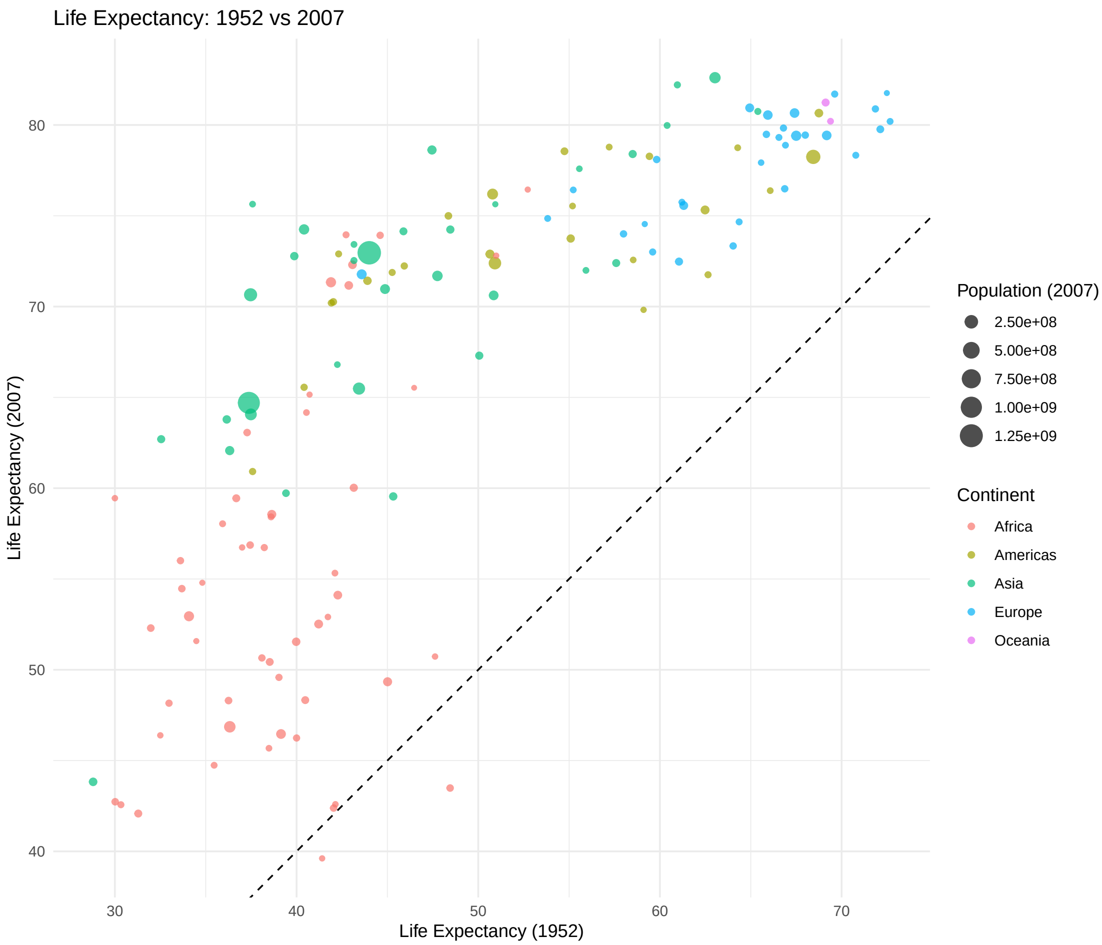
African countries like Gabon, Botswana, and Equatorial Guinea also appeared in the top three for the continent, though their GDP per capita was still lower compared to wealthier regions.

2.2 Development Progress Visualization

Create a scatter plot comparing each country's 1952 vs. 2007 life expectancy.

```r
library(dplyr)
library(ggplot2)

lifeExp_1952 <- gapminder %>%
  filter(year == 1952) %>%
  select(country, continent, lifeExp_1952 = lifeExp)

lifeExp_2007 <- gapminder %>%
  filter(year == 2007) %>%
  select(country, lifeExp_2007 = lifeExp, pop_2007 = pop)

lifeExp_data <- left_join(lifeExp_1952, lifeExp_2007, by = "country")

lifeExp_data <- lifeExp_data %>%
  filter(!is.na(lifeExp_1952), !is.na(lifeExp_2007))
```

```r
# Create the scatter plot
ggplot(lifeExp_data,
       aes(x = lifeExp_1952, y = lifeExp_2007,
           color = continent, size = pop_2007)) +
  geom_point(alpha = 0.7) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  coord_equal() +
  labs(
    title = "Life Expectancy: 1952 vs 2007",
    x = "Life Expectancy (1952)",
    y = "Life Expectancy (2007)",
    color = "Continent",
    size  = "Population (2007)"
  ) +
  theme_minimal() +
  theme(legend.position = "right")
```




```
# Your code here - add the diagonal reference line using geom_abline()
# Add points with color by continent and size by population
# Add professional labels and theme
```

Analysis: The plot shows that most countries saw clear improvements in life expectancy from 1952 to 2007, since nearly all points are above the diagonal line. While Europe and parts of Asia reached the highest levels, many African countries still lagged behind, though they also made progress.

2.3 Development Progress Classification

Using the `lifeExp_data` from Task 2.1.2, classify countries by improvement level.

```r
library(dplyr)

lifeExp_data <- lifeExp_data %>%
  mutate(
    improvement = lifeExp_2007 - lifeExp_1952,
    improvement_level = case_when(
      improvement >= 30 ~ "Major Improvement",
      improvement >= 15 ~ "Moderate Improvement",
      improvement >= 0  ~ "Minor Improvement",
      TRUE              ~ "Decline"
    )
  )

improvement_summary <- lifeExp_data %>%
  group_by(continent, improvement_level) %>%
  summarise(n_countries = n(), .groups = "drop")

print(improvement_summary, n = Inf)
```

```
# A tibble: 13 x 3
   continent improvement_level    n_countries
   <fct>     <chr>                      <int>
 1 Africa    Decline                        2
 2 Africa    Major Improvement              1
 3 Africa    Minor Improvement             23
 4 Africa    Moderate Improvement          26
 5 Americas  Major Improvement              1
 6 Americas  Minor Improvement              8
 7 Americas  Moderate Improvement          16
 8 Asia      Major Improvement              7
 9 Asia      Minor Improvement              2
10 Asia      Moderate Improvement          24
11 Europe    Minor Improvement             23
12 Europe    Moderate Improvement           7
13 Oceania   Minor Improvement              2
```

2.3.1 Analysis Questions

1. Which continent has the most countries with "Major Improvement"?

```r
# Your code here - analyze major improvement by continent
major_improvement_summary <- lifeExp_data %>%
  filter(improvement_level == "Major Improvement") %>%
  group_by(continent) %>%
  summarise(n_countries = n(), .groups = "drop") %>%
  arrange(desc(n_countries))

print(major_improvement_summary)
```

```
# A tibble: 3 x 2
  continent n_countries
  <fct>           <int>
1 Asia                7
2 Africa              1
3 Americas            1
```

Asia is the continent with most "Major Improvement Countries".

2. Are there any countries that experienced decline? Which ones and why might this have happened?

```r
# Your code here - find and examine countries with decline
decline_countries <- lifeExp_data %>%
  filter(improvement_level == "Decline") %>%
  select(country, continent, lifeExp_1952, lifeExp_2007, improvement)

print(decline_countries, n = Inf)
```

```
# A tibble: 2 x 5
  country   continent lifeExp_1952 lifeExp_2007 improvement
  <fct>     <fct>            <dbl>        <dbl>       <dbl>
1 Swaziland Africa            41.4         39.6       -1.79
2 Zimbabwe  Africa            48.5         43.5       -4.96
```

In Swaziland and Zimbabwe, life expectancy fell mainly because of the HIV/AIDS epidemic. Economic crisis and weak health systems also made the situation worse, preventing improvements seen in other countries.

3. What does this tell us about global health convergence?

The results suggest that global health is partly converging. Many countries in Asia and the Americas made large gains in life expectancy, helping them catch up with Europe. However, Africa shows a mixed picture, with most countries improving only moderately and a few even declining, which means convergence is uneven across regions.

2.4 Economic Development Trajectories

Create a line plot showing life expectancy changes over time for selected countries.

2.4.1 Step 1: Guided Discovery

```
life_exp_changes <- lifeExp_data %>%
  mutate(change = lifeExp_2007 - lifeExp_1952) %>%
  select(country, continent, lifeExp_1952, lifeExp_2007, change)

top_gainer <- life_exp_changes %>%
  arrange(desc(change)) %>%
  slice(1)

decliners <- life_exp_changes %>%
  filter(change < 0)

print(top_gainer)
```

```
# A tibble: 1 x 5
  country continent lifeExp_1952 lifeExp_2007 change
  <fct>   <fct>            <dbl>        <dbl>  <dbl>
1 Oman    Asia              37.6         75.6   38.1
```

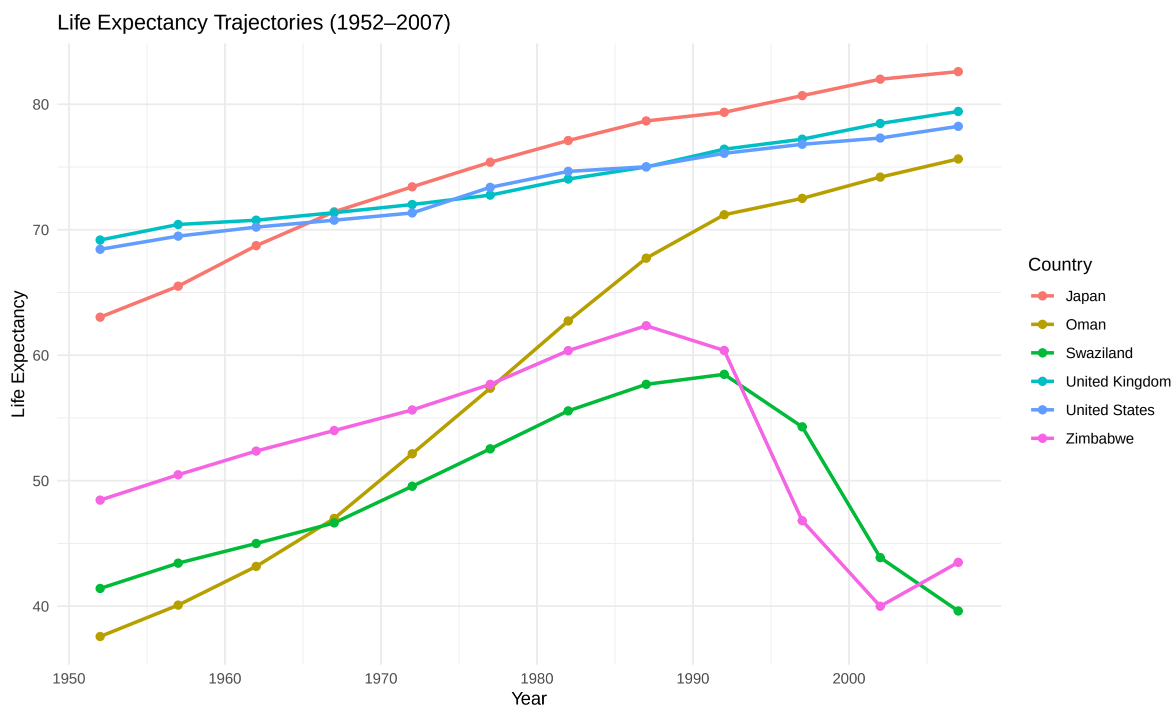
```
print(decliners)
```

```
# A tibble: 2 x 5
  country   continent lifeExp_1952 lifeExp_2007 change
  <fct>     <fct>            <dbl>        <dbl>  <dbl>
1 Swaziland Africa            41.4         39.6  -1.79
2 Zimbabwe  Africa            48.5         43.5  -4.96
```

2.4.2 Step 2: Create Trajectory Visualization

```
# Step 2: Create Trajectory Visualization
selected_countries <- c("Oman", "Swaziland", "Zimbabwe",
                        "Japan", "United States", "United Kingdom")
```

```r
trajectory_data <- gapminder %>%
  filter(country %in% selected_countries)

ggplot(trajectory_data, aes(x = year, y = lifeExp, color = country)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  labs(title = "Life Expectancy Trajectories (1952-2007)",
       x = "Year",
       y = "Life Expectancy",
       color = "Country") +
  theme_minimal()
```



Caption: I was most surprised by the United States and the United Kingdom. In 1952, they already had some of the highest life expectancies in the world (around 69 and 68 years), but by 2007 they had only risen to about 78–79 years. Meanwhile, countries like Japan, Singapore, and Hong Kong made much larger gains and ended up surpassing them.

3 Research Discovery & Critical Analysis

3.1 Choose Your Research Question

Selected Research Question: *[A, B, C, D, or E]*

[A]

3.2 Research Analysis

3.2.1 Data Analysis

```
mid_income_1987 <- gapminder %>%
  filter(year == 1987, gdpPercap >= 3000, gdpPercap <= 12000) %>%
  select(country, continent, gdpPercap_1987 = gdpPercap)

mid_income_status <- gapminder %>%
  filter(year == 2007, country %in% mid_income_1987$country) %>%
  select(country, continent, gdpPercap_2007 = gdpPercap)

mid_income_trap <- left_join(mid_income_1987, mid_income_status, by = c("country", "continen
  filter(gdpPercap_2007 < 12000)

print(mid_income_trap)
```

```
# A tibble: 37 x 4
   country                continent gdpPercap_1987 gdpPercap_2007
   <fct>                  <fct>              <dbl>          <dbl>
 1 Albania                Europe             3739.          5937.
 2 Algeria                Africa             5681.          6223.
 3 Bosnia and Herzegovina Europe             4314.          7446.
 4 Brazil                 Americas           7807.          9066.
 5 Bulgaria               Europe             8240.         10681.
 6 Colombia               Americas           4903.          7007.
 7 Congo, Rep.            Africa             4201.          3633.
 8 Costa Rica             Americas           5630.          9645.
 9 Cuba                   Americas           7533.          8948.
10 Ecuador                Americas           6482.          6873.
# i 27 more rows
```

3.2.2 Visualization

```r
# Your code here - create one clear plot that supports your findings
ggplot(mid_income_trap, aes(x = gdpPercap_1987, y = gdpPercap_2007, label = country, color =
  geom_point(size = 3) +
  geom_text(vjust = -0.5, size = 3) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  labs(title = "Middle-Income Trap (1987-2007)",
       x = "GDP per Capita in 1987",
       y = "GDP per Capita in 2007") +
  theme_minimal()
```

3.2.3 Interpretation (150-200 words)

This analysis shows that many countries reached middle-income status by 1987 but failed to become high-income by 2007. Latin America is the clearest case, where growth slowed despite earlier progress. This suggests that the middle-income trap is linked to structural issues such as weak institutions, lack of innovation, or dependence on commodities. For example, several Latin American countries remained stuck because they could not upgrade their industries or reduce inequality. While this analysis only looks at GDP per capita and misses social or political factors, it highlights how hard it is to sustain growth once basic industrialization is achieved.